

ECON G6411-A, Fall 2026
Introduction to Econometrics
Columbia University

Professor: Serena Ng
Class: M, W. 10.10-11.25am (337 Mudd Hall)
Office Hours: Th. 9:30-11am (IAB 1126D)

Teaching Assistant: Jesse Chen
Recitation: T, 8-9:30am (Hamilton 503)
Office Hours: W, 2-3pm (Lehman Library Group Study Room)

Pre-requisites EC 6411 is the first course in the Ph.D sequence in Econometrics. The goal is to provide you with a solid foundation of econometric methods and their use in practice. Part A consists of 13 classes (September 9 to October 21, 2026). Students should have taken (i) an introductory econometrics course based on the textbook by Stock-Watson, Wooldridge, Greene, or equivalent that includes a basic knowledge of probability and statistics; (ii) a university level course in Linear Algebra. Knowledge of basic calculus would be desirable.

Evaluation:

- (i) (10%): 2 problem sets
- (ii) (15%): 30 minute closed-books/closed-notes exam during recitation on Sept. 22, 2026
- (iii) (15%): 30 minute exam closed-book/closed-notes during recitation on October 6, 2026
- (iv) (60%): 90 minute closed-book/closed-notes final exam on Friday October 23 2026.

*A make-up exam will carry a penalty of 20 (out of 60) points. There will be no exception.

Useful References :

Cameron, B. and P. Trevedi. *Microeconometrics*, 2002, Cambridge University Press.
Hansen, B. *Econometrics*, 2022, Princeton University Press.
Hayashi, F. *Econometrics*, 2000, Princeton University Press.
Stock, J. and W. Watson *Introduction to Econometrics*, 4th edition, Pearson.

AI Policy You are free to use AI to help you learn. However, the use of generative AI tools (such as ChatGPT, Claude, Gemini, GitHub Copilot) *and* web-searches are strictly prohibited during exams. Answers based on AI and internet resources in exams and problem sets will be treated as unauthorized assistance and/or plagiarism. See guidelines from the Office of the Provost on Generative AI and Academic Integrity:
<https://provost.columbia.edu/content/office-senior-vice-provost/ai-policy> and
<https://cssi.columbia.edu/content/academic-violations>.

Topics

1. Linear regression model (4 classes, Exam 1):
 - Algebraic properties: Hansen 3.1-3.18
 - Finite sample properties: Hansen 4.1-4.18, 9.1-9.9
 - Large sample properties: Hansen 6, 7.1-7.17, 9.15-9.18
 - Ridge and Lasso. Hansen 29.5, 29.8
 - Maximum likelihood: CT: 5.6., 5.7.
2. Hypothesis testing and Model Selection (4 classes, Exam 2):
 - Wald test of linear hypothesis, CT: 7.2, Hansen 9.9
 - Likelihood ratio test, CT 7.3, Hansen 9.11
 - Lagrange multiplier test, CT 7.3.5, Hansen 9.16
 - Delta method and non-linear hypothesis, CT 7.2.8
 - AIC/BIC, Hansen 28.3, 28.4
 - Cross-validation Hansen 28.8
 - Hausman test, CT 8.3, Hansen 9.15
 - Non-nested tests, CT 8.5.3
3. Bayesian Estimation (3 classes):
 - Theil-Goldberger mixed estimation, CT 13.3.3
 - Likelihood principle, Bayes rule, CT 5.6.1, CT 13.1
 - Prior and posterior distributions, CT 13.2
 - Conjugate analysis for the normal linear model, CT 13.3
4. Time Series Regression and Forecasting (2 classes):
 - Stationarity, SW 15.2, 15.7
 - Autocorrelation autoregression, SW 15.1, SW 15.3
 - Autoregressive distributed lag models, SW 15.4
 - Dynamic causal effects SW 16.2, 16.3, 16.5
 - HAC and HAR standard errors, SW 16.4
 - Vector-autoregression, SW. 17.1